CHAYAN SEHGAL

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Professional Summary

Software Engineer with 1+ year of professional experience and hands-on backend/API development using Node.js, Express.js, FastAPI, Python, SQL, and MongoDB. Experienced in building RESTful APIs, integrating external services, authentication and payment workflows, debugging production issues, and developing maintainable application components. Strong foundation in OOP, data structures, databases, API integration, testing, Git, and Agile development, with additional experience in AI-assisted development and LLM applications.

Technical Skills

Languages	Java, Python, JavaScript (ES6+), TypeScript, SQL
Backend	Node.js, Express.js, FastAPI, RESTful APIs, API Integration, JWT Authentication, JSON, HTTP
Databases	MySQL, MongoDB, ChromaDB, Mongoose, SQL Queries, Data Modeling
Software Engineering	OOP, Data Structures & Algorithms, DRY, Debugging, Unit Testing, Edge Cases, Git, GitHub, Postman
Integration	Razorpay, REST APIs, Third-Party Services, Webhooks, HMAC-SHA256 Signature Verification
AI / AI-Assisted Development	LangChain, Multimodal RAG, Gemini API, Groq API, Hugging Face, Embeddings, Vector Search, Prompt Engineering
Web	React.js, Next.js, HTML5, CSS3, Tailwind CSS, Vite
DevOps / Process	GitHub Actions, CI/CD, Vercel, Render, Agile/Scrum, JIRA

Professional Experience

Junior Software Engineer — Innova Solutions Aug 2025 – Present
Hyderabad, India

- Contributed to TalentX, a production recruitment platform, delivering end-to-end application workflows and API-integrated features with frontend and backend teams.
- Integrated and maintained REST API workflows, handling request/response data, frontend-backend communication, error states, and application-level debugging.
- Diagnosed application bottlenecks and production issues through debugging and profiling; reduced page load time from 3.2s to 2.1s (34%).
- Built reusable TypeScript components and shared application patterns across 3 product repositories, applying DRY and maintainability principles.
- Implemented GitHub Actions CI checks for linting and formatting, contributing to zero rollbacks across 6 sprint cycles.

Software Development Engineer Intern — Innova Solutions Oct 2024 – Jul 2025
Hyderabad, India

- Delivered 12+ application features and resolved 20+ JIRA tickets in a 9-member Agile team while collaborating with developers and QA.
- Investigated application defects, reproduced issues, tested fixes, and contributed to improving software reliability and performance.
- Profiled high-traffic application flows and improved Largest Contentful Paint from 4.8s to 2.1s (56%) through targeted performance optimization.

Backend & AI Projects

Rivanto Mattress – Secure E-commerce Platform Next.js, Node.js, TypeScript, REST APIs, Razorpay

- Developed backend-integrated shopping workflows and REST/API interactions for product, checkout, and payment flows.
- Implemented Razorpay payment verification using HMAC-SHA256 signatures, timing-safe comparison, and server-side order validation.
- Built maintainable application workflows with server-side validation and secure third-party payment integration.

Talkify – Real-Time Communication Hub Node.js, Express.js, MongoDB, Socket.IO

- Built backend APIs with Node.js and Express.js and integrated MongoDB for user and application data.
- Implemented Socket.IO-based real-time messaging, typing presence, online-status indicators, and concurrent chat-room communication.
- Designed API-driven application workflows with a focus on reliable real-time communication and maintainable backend code.

Multimodal RAG System Python, LangChain, ChromaDB, Hugging Face

- Developed Python document-processing and retrieval pipelines for semantic search and context-aware LLM responses.
- Implemented embeddings, vector storage, multi-query retrieval, document ranking, context selection, and Reciprocal Rank Fusion (RRF).

Education

Chitkara University – B.E. in Computer Science CGPA: 9.31/10.0 — Top 5% of Graduating Class